CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

5.2 Gravitational Potential Energy & Kinetic Energy

Syllabus Section: 5. Work, Energy & Power **Paper Type:** Multiple Choice (Paper 1) **Total:** 31 Questions (31 Marks)

Question 1 Easy [1 mark]

What is the approximate kinetic energy of an Olympic athlete when running at maximum speed during a 100m race?

A 400 J

B 4000 J

C 40 000 J

D 400 000 J

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- In the first topic we looked at how we can estimate different quantities
- Here we need to estimate the maximum speed and mass of an Olympic athlete
- A reasonable estimate for the time taken to complete a 100 m race in an Olympic competition is about 10 s
- Let the maximum speed of the Olympic athlete be $v = \frac{100}{10} = 10 \quad \text{m} \quad \text{s}^{-1}$
- Let the mass of the Olympic athlete be about 80 kg
  - Kinetic energy $\frac{1}{2}mv^2 = \frac{1}{2} \times 80 \times 10^2 = 4000 \quad \text{J}$

Question 2 Easy [1 mark]

A body travelling with a speed of 10 m s^{-1} has kinetic energy 1500 J.

If the speed of the body is increased to 40 m s^{-1}, what is its new kinetic energy?

A 4.5 kJ

B 6 kJ

C 24 kJ

D 1350 kJ

Mark Scheme & Explanation:

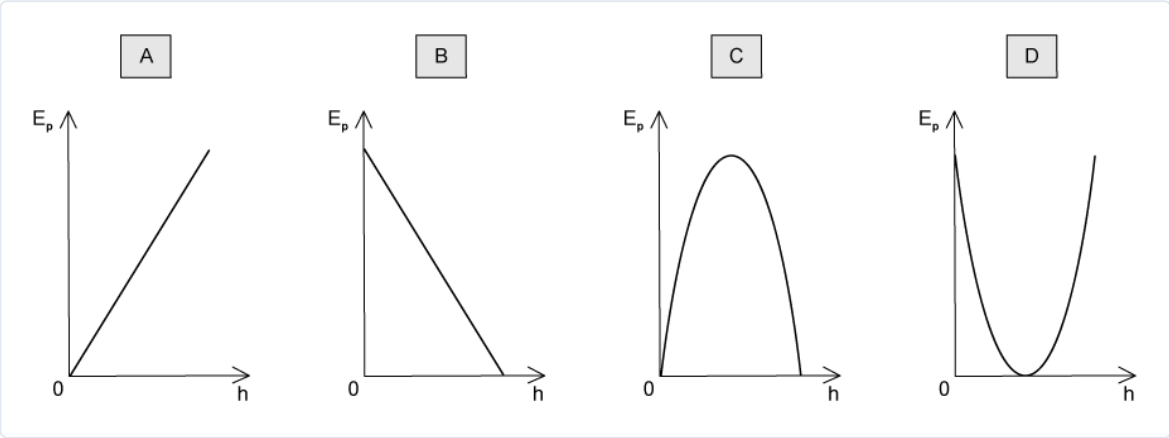
*The correct answer is **C** because:*

- The kinetic energy of an object is equal to
  - $E_k = \frac{1}{2}mv^2$
- The mass of the object is constant, but we must know its value to find the new kinetic energy. Therefore:
- $1500 = \frac{1}{2}m \times 10^2$
- $m = \frac{2 \times 1500}{100} = 30 \quad \text{kg}$
- Now we have our m and v values, so we can work out the new kinetic energy :
  - $E_k = \frac{1}{2} \times 30 \times 40^2 = 24 \quad 000 \quad \text{J}$
- The new kinetic energy is 24 kJ

Question 3 Easy [1 mark]

An object is thrown into the air.

Which graph shows how the potential energy E_p of the object varies with height h above the ground?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The height above the ground is Δh
- So, as h increases, the position of the object above the ground gets higher
- Potential energy is equal to $\Delta E_p = mg\Delta h$
- Therefore, as h increases, potential energy E_p increases linearly since the relationship between ΔE_p and Δh is directly proportional $\Delta E_p \quad \alpha \quad \Delta h$

B is incorrect because this graph shows the object would have more GPE on the ground than when it is a height h above the ground

C & D are incorrect because these graphs show a non-linear increase/decrease in GPE as h increases

Question 4 Easy [1 mark]

A car of mass 1000 kg travels forward at 25 m s^{-1} and then backwards at 5 m s^{-1}

What is the magnitude of the change in the kinetic energy of the car?

A 200 kJ

B 300 kJ

C 325 kJ

D 450 kJ

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The change in kinetic energy is equal to
  - $\Delta E_k = \frac{1}{2}mu^2 - \frac{1}{2}mv^2$
- Where u = 25 m s^{-1} and v = 5 m s^{-1}
  - $\Delta E_k = \frac{1}{2} \times 100025^2 \quad - 5^2 = 300 \quad 000 \quad \text{J}$
- The change in kinetic energy is 300 kJ

Question 5 Easy [1 mark]

To get to his office from the entrance of the building, a man has to walk up six flights of stairs.

The height of each flight is 2.5 m and the man has a mass of 80 kg.

What is the approximate gain in the man's gravitational potential energy during the climb?

A 1200 J

B 2000 J

C 4800 J

D 12 000 J

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Gravitational potential energy (GPE) $\Delta E_p = mg\Delta h$
- The GPE gained by walking up one flight of stairs is $\Delta E_p = 80 \times 9.8 \times 2.5 = 1960 \text{ J}$
- The GPE gained by walking up six flights of stairs is $\Delta E_p = 6 \times 1960 = 11\ 760 \text{ J}$
    - Therefore, the man gains 12 000 J in climbing six flights of stairs

Question 6 Easy [1 mark]

A car with a total mass of 1400 kg is travelling at 108 km h^{-1}

What is the kinetic energy of the car?

A 21 kJ

B 42 kJ

C 630 kJ

D 8160 kJ

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- The speed is in km h^{-1} so we must first convert it to m s^{-1}
  - $108 \quad \text{km} \quad \text{h}^{-1} = \frac{108 \times 1000}{60 \times 60} = 30 \quad \text{m} \quad \text{s}^{-1}$
- The kinetic energy is therefore equal to
  - $E_k = \frac{1}{2}mv^2 = \frac{1}{2} \times 1400 \times 30^2 = 630 \quad 000 \quad \text{J}$
- The kinetic energy of the car is 630 kJ

Question 7 Easy [1 mark]

The formula for kinetic energy is $E_k = \frac{1}{2}mv^2$

Which kinematic equation is used in the derivation of this formula?

A $v = u + at$

B $s = \frac{u+v}{2} \quad t$

C $s = ut + \frac{1}{2}at^2$

D $v^2 = u^2 + 2as$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- In the derivation of kinetic energy (KE), we start by equating KE with the work done by the moving body, then add in Newton's second law of motion $F = ma$
  - $\Delta E_k = W = F\Delta s = ma\Delta s$
- Then we want to use a kinematic equation which contains the term $a\Delta s$
  - $v^2 = u^2 + 2a\Delta s$
  - $a\Delta s = \frac{v^2 - u^2}{2}$
  - $\Delta E_k = ma\Delta s = m \quad \frac{v^2 - u^2}{2}$
- Now if we expand this equation, we get
  - $\Delta E_k = \frac{mv^2}{2} - \frac{mu^2}{2}$
- We assume that the object started at rest and accelerated with a uniform rate of *a* up to speed *v,* therefore *u* = 0 and this leaves us with
  - $E_k = \frac{1}{2}mv^2$

Question 8 Easy [1 mark]

The formula for gravitational potential energy is $\Delta E_p = mg\Delta h$

From which concept is this equation derived?

A Work Done = Force × Displacement

B Force = Mass × Acceleration

C Power = Force × Speed

D Distance = Speed × Time

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- In the derivation of gravitational potential energy (GPE), we start with the definition of work done
  - $Work\ \ done = force \times displacement = F\Delta s$
- Next, we consider the case of an object being raised by a height Δh
- The only force acting on this object will be its weight
  - $F = W = mg$
- To move an object against gravity, the force applied upwards must equal the downwards force gravity exerts on the object, mg. So, if the object is moved against gravity by a distance Δh the work done is given by :
  - $\Delta E_p = mg\Delta h$

Question 9 Easy [1 mark]

A cyclist is travelling at a constant speed up a hill. The frictional force resisting the cyclist's motion is 8.0 N.

The cyclist uses 450 J of energy to travel 20 m.

What is the increase in gravitational potential energy of the cyclist?

A 160 J

B 290 J

C 440 J

D 610 J

Mark Scheme & Explanation:

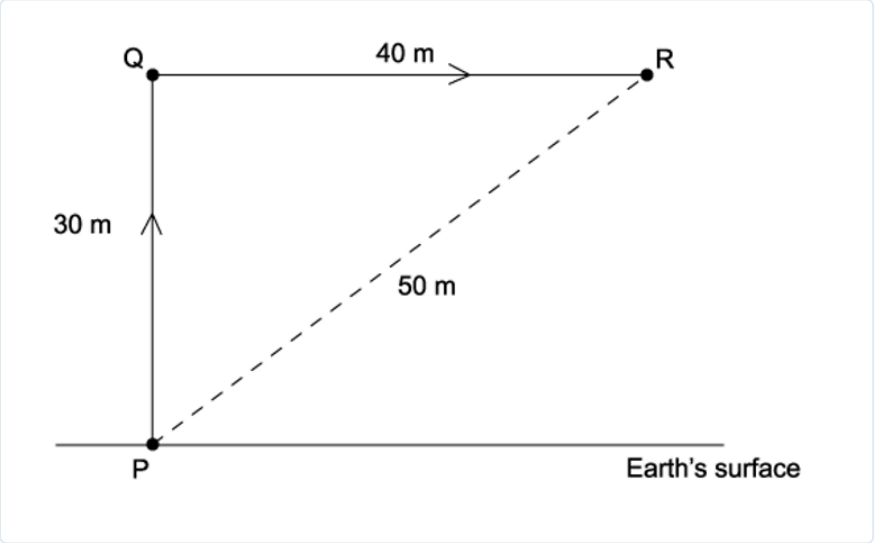
*The correct answer is **B** because:*

- Since the cyclist is moving at a constant speed, he is not gaining any kinetic energy
- Work done against friction $W = Fs = 8 \times 20 = 160 \quad \text{J}$
- From the conservation of energy :
- Work done by cyclist = Gain in GPE + Work done against friction
    - $450 = \Delta E_p + 160$
- So, the ΔE_p gained by the cyclist is
    - $\Delta E_p = 450 - 160 = 290$ J

Question 10 Easy [1 mark]

A stone of weight 4.0 N in the Earth's gravitational field is moved from P to Q and then to R along the path shown.

How much potential energy does the stone gain?



A 120 J

B 200 J

C 280 J

D 1200 J

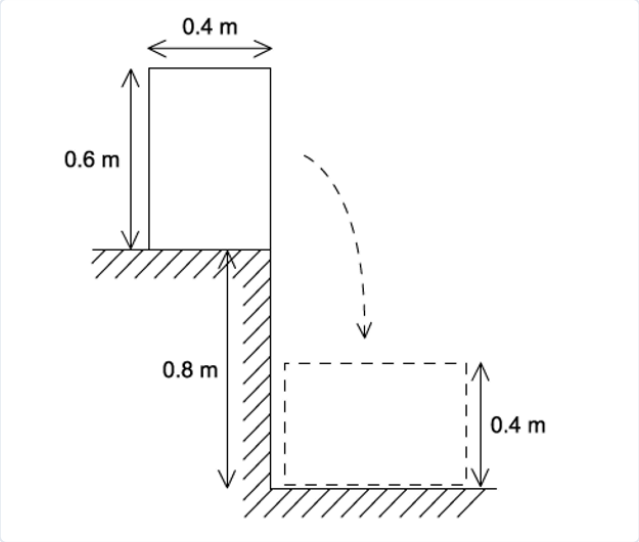
Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Gravitational potential energy $\Delta E_p = mg\Delta h$
- *h* refers to the vertical distance travelled by the stone, which is 30 m and *mg* refers to the weight of the stone, which is 4 N
    - Therefore, $\Delta E_p = 4 \times 30 = 120$ J

Question 11 Medium [1 mark]

A uniform solid block has weight 500 N, width 0.4 m and height 0.6 m. The block rests on the edge of a step of depth 0.8 m, as shown.



The block is knocked over the edge of the step and rotates through 90° before coming to rest with the 0.6 m edge horizontal.

What is the change in gravitational potential energy of the block?

A 300 J

B 400 J

C 450 J

D 550 J

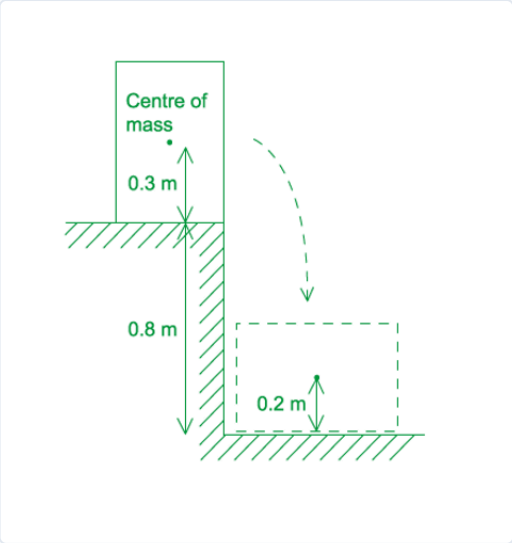
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- In this question we need to take the change in height Δh from the centre of mass of the block
- The centre of mass is the point where the force of gravity is taken to act. We need to consider the centre of mass, else each point on the box would have its own GPE as they have different heights
- We are told the block is uniform, so the centre of mass can be considered to be at its centre
- At the top of the step:
    - Height of position of centre of mass above ground

$h_i = 0.8 + \frac{0.6}{2} = 0.8 + 0.3 = 1.1 \text{ m}$

- At the bottom of the step:
  - Height of position of centre of mass above ground $h_f = \frac{0.4}{2} = 0.2 \text{ m}$
- Change in GPE = final GPE – initial GPE
  - $\Delta E_p = mg\Delta h = 500 \times 1.1 - 0.2 = 450 \text{ J}$
- Therefore, the loss in GPE is 450 J



Question 12 Medium [1 mark]

A body travelling with a speed of 20 $\mathrm{m\ \ s^{-1}}$ has kinetic energy E_k

If the speed of the body is increased to 80 $\mathrm{m\ \ s^{-1}}$, what is its kinetic energy?

A $4E_k$

B $8E_k$

C $12E_k$

D $16E_k$

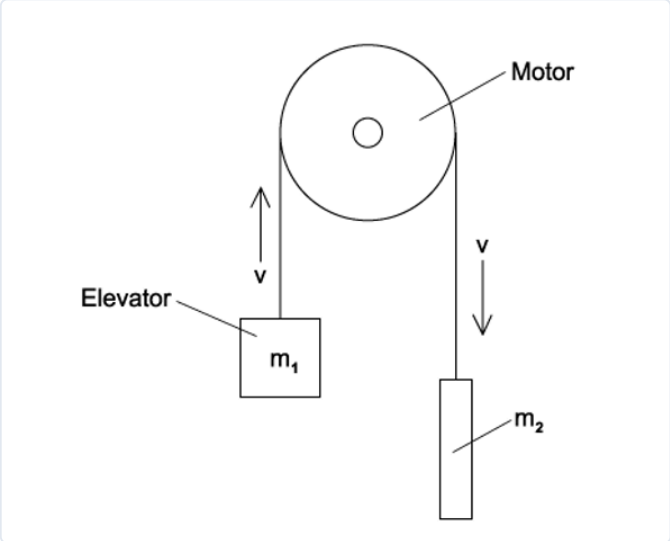
Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Initial kinetic energy $E_k = \frac{1}{2}mv^2 = \frac{1}{2}m20^2$
- Final kinetic energy $= \frac{1}{2}mv^2 = \frac{1}{2}m80^2$
- We need to write the final kinetic energy in terms of E_k
- The ratio of final KE to initial KE is $= \frac{\frac{1}{2}m80^2}{\frac{1}{2}m20^2} = 16$
    - Therefore, final KE $= 16 \ \ E_k$

Question 13 Medium [1 mark]

The diagram shows a lift system in which the elevator (mass m_1) is partly counterbalanced by a heavy weight (mass m_2)



At what rate does the motor provide energy to the system when the elevator is rising at a steady speed v ? (g = acceleration of free fall)

A $\frac{1}{2}m_1 v^2$

B $\frac{1}{2}m_1 - m_2 v^2$

C $m_1 gv$

D $m_1 - m_2 gv$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The rate at which the motor provides energy is the power of the motor
- In the lift system, the mass m_2 works downwards under gravity, causing the elevator of mass m_1 to move up. However, the motor also provides some energy in moving the elevator up
- The gravitational potential energy of the masses is changing continuously and at a steady rate, since they are moving at a steady speed
- GPE of elevator = $m_1 gh_1$
    - Where h_1 is its height above the ground

- Rate of change of GPE of elevator = $\frac{m_1 g h_1}{t} = m_1 g v$
- GPE of heavy weight = $m_2 g h_2$
    - Where h_2 is its height above the ground
- Rate of change of GPE of heavy weight = $\frac{m_2 g h_2}{t} = m_2 g v$
- Let the power of the motor be *P*
    - $m_1 g v = m_2 g v + P$
    - $P = m_1 - m_2 g v$
- Therefore, the power of the motor is equal to $m_1 - m_2 g v$

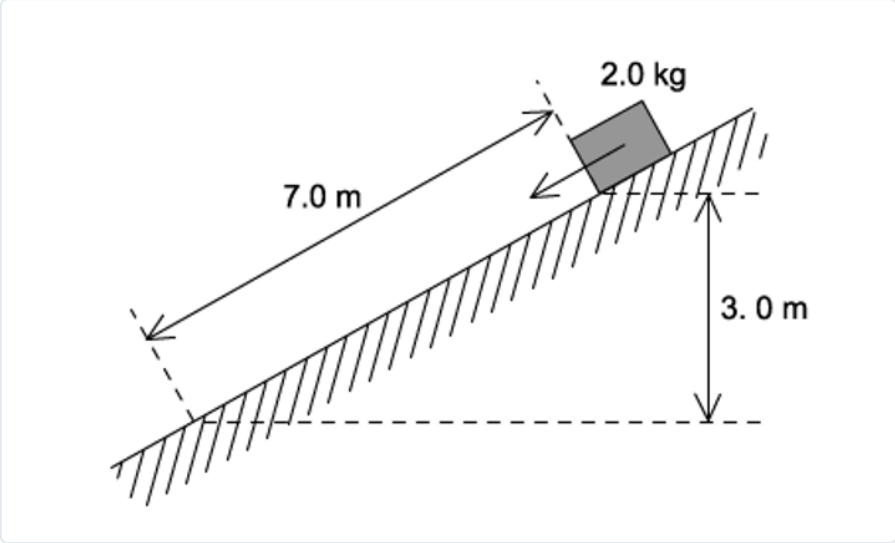
A & B are incorrect because these choices have units of energy, not power. If **B** was correct, then there would have no need for the heavy weight

C is incorrect because this option ignores the influence of the other mass m_2

Question 14 Medium [1 mark]

A block of mass 2.0 kg is released from rest on a slope. It travels 7.0 m down the slope and falls a vertical distance of 3.0 m

The block experiences a frictional force parallel to the slope of 5.0 N.



What is the speed of the block after falling this distance?

A 4.9 m s^{-1}

B 6.6 m s^{-1}

C 8.6 m s^{-1}

D 10.1 m s^{-1}

Mark Scheme & Explanation:

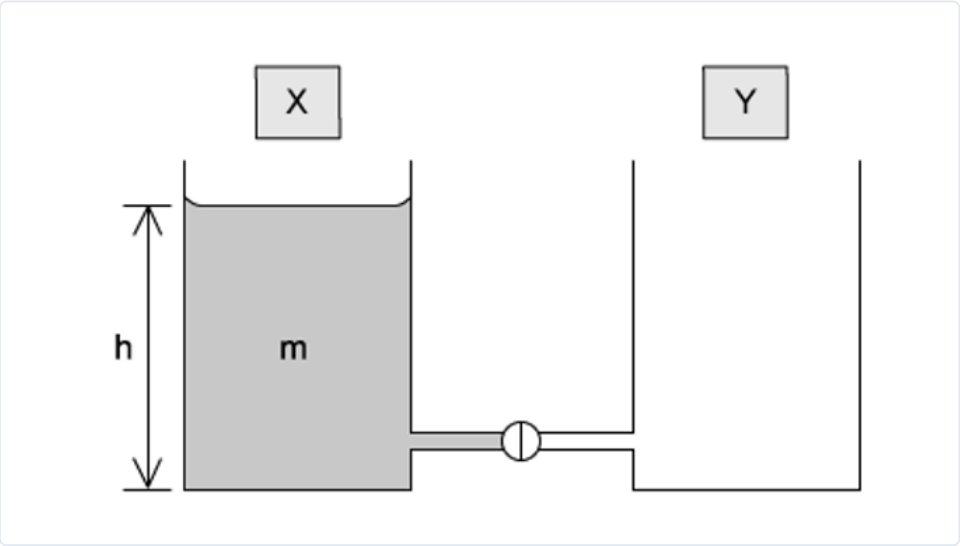
*The correct answer is **A** because:*

- At the top of the slope, the block has gravitational potential energy (GPE). As the block falls, some GPE is converted into kinetic energy of the block and work done due the frictional force
- ΔGPE = KE + Work done against friction
- The loss in height Δ*h* of the block is 3.0 m
  - $\Delta E_p = mg\Delta h = 2.0 \times 9.81 \times 3.0 = 58.86 \quad \text{N}$
- Frictional force = 5.0 N

- Distance moved against friction = 7.0 m
    - Work done against friction $Fs = 5.0 \times 7.0 = 35.0 \text{ J}$
- ΔGPE = KE + Work done against friction
    - $58.86 = E_k + 35.0$
- KE of block at bottom = $58.86 - 35.0 = 23.86 \text{ J}$
    - $\frac{1}{2}mv^2 = 23.86$
    - $v = \sqrt{\frac{23.86 \times 2}{2.0}} = 4.88 \text{ m s}^{-1}$
- Therefore, the speed of the block after falling is 4.9 m s^{-1}

Question 15 Medium [1 mark]

The diagram shows two identical vessels X and Y connected by a short pipe with a tap.



Initially, X is filled with water of mass *m* to a depth *h*, and Y is empty.

When the tap is opened, water flows from X to Y until the depths of water in both vessels are equal.

How much potential energy is lost by the water during this process? (*g* = acceleration of free fall)

A 0

B $\frac{mgh}{4}$

C $\frac{mgh}{2}$

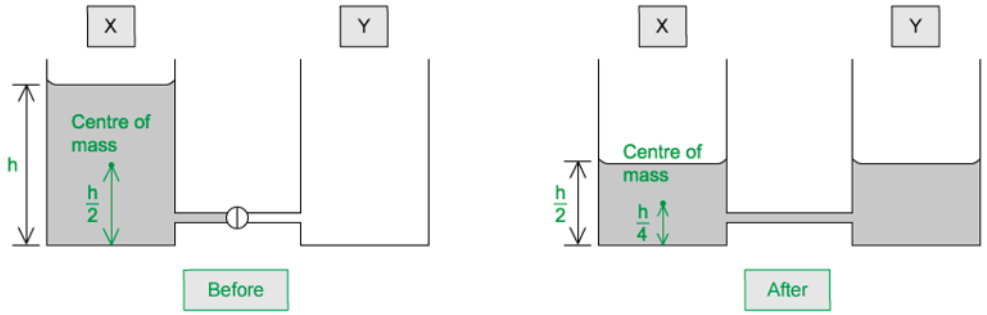
D mgh

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Consider the water in both vessels as one body with one mass
- Since the mass of the water is spread all over the volume it occupies, the centre of mass should be considered when calculating the gravitational potential energy (GPE)
- Before the tap is opened, the centre of mass of the water in vessel X may be considered to be at its centre which is at half the water level $\frac{h}{2}$
    - Potential energy of water before tap is opened = $mg\frac{h}{2} = \frac{mgh}{2}$

- When the tap is opened the total mass of water in the whole body is the same, but the water level is now at a height of $\frac{h}{2}$ in both vessels
- The centre of mass may be considered to be at the centre of the volume of water, which is, again, at a height of half the water level so the centre of gravity is now at a height of $\frac{h}{4}$
    - Potential energy of water after tap has been opened = $mg\frac{h}{4} = \frac{mgh}{4}$
- Loss in potential energy$\frac{mgh}{2}$ - $\frac{mgh}{4} = \frac{mgh}{4}$



Question 16 Medium [1 mark]

A concrete cube of side 0.60 m and uniform density 2.0×10^3 kg m^{-3} is lifted 5.0 m vertically by a crane.

What is the change in potential energy of the cube?

A 0.75 kJ

B 7.4 kJ

C 21 kJ

D 470 kJ

Mark Scheme & Explanation:

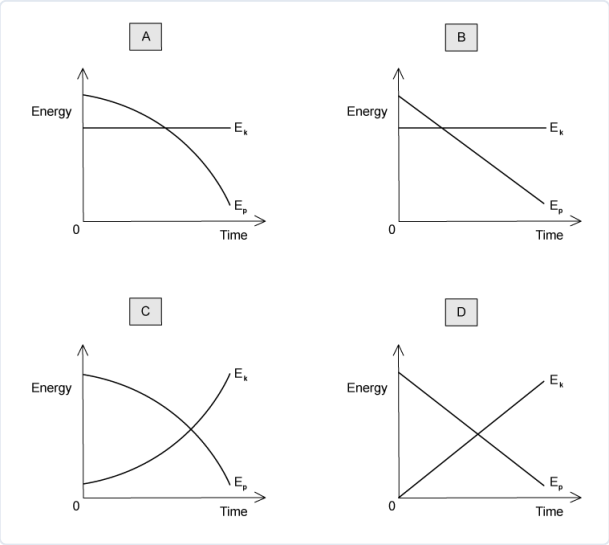
*The correct answer is **C** because:*

- The density of the cube is given as ρ = 2.0×10^3 kg m^{-3}
- The volume of the cube is given as $V = a^3 = 0.6^3$
- Density is equal to $\rho = \frac{m}{V}$
  - So, mass is equal to $m = \rho \times V = 2.0 \times 10^3 \times 0.6^3 = 432 \text{ kg}$
- Now, we have to find the change in potential energy of the cube. At ground level, the potential energy is equal to zero and the potential energy at height *h* is given by
  - $\Delta E_p = mg\Delta h = 432 \times 9.81 \times 5.0 = 21\ 190 \text{ J}$
- Therefore, the change in GPE is 21 kJ

Question 17 Medium [1 mark]

A steel ball is falling at constant speed in oil.

Which graph shows the variation with time of the gravitational potential energy E_p and the kinetic energy E_k of the ball ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Kinetic energy (KE) is equal to $E_k = \frac{1}{2}mv^2$
- As the speed it constant, KE also remains constant
  - On the graph, this will be shown as a straight, horizontal line
- Gravitational potential energy (GPE) is equal to $\Delta E_p = mg\Delta h$
- As the speed is constant, the rate at which the distance *h* decreases is also constant, so the decrease in GPE is also constant
  - On the graph, this will be shown as a straight, diagonal line starting from a maximum value with a negative gradient

A & C are incorrect because the speed, and hence the change in distance, of the steel ball is constant, meaning the relationship between these quantities and E_k and ΔE_p will be linear which would not result in curved lines on the graph

D is incorrect because this graph would be accurate for an object accelerating towards the ground, however the ball is travelling at a constant speed so the KE must be constant

Question 18 Medium [1 mark]

Car X is travelling at half the speed of car Y. Car X has twice the mass of car Y.

Which statement is correct?

A car X has half the kinetic energy of car Y

B car X has one quarter of the kinetic energy of car Y

C car X has twice the kinetic energy of car Y

D the two cars have the same kinetic energy

Mark Scheme & Explanation:

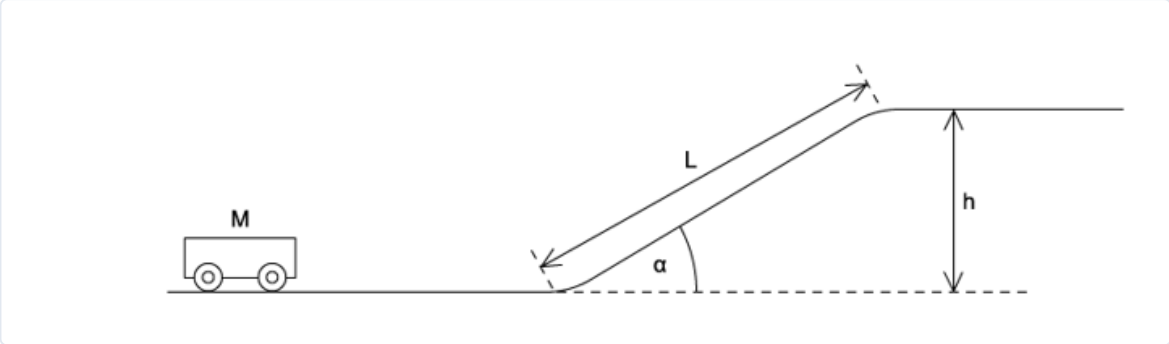
*The correct answer is **A** because:*

- Let the speed of car X = v
- Let the mass of car X = $2m$
- Let the speed of car Y = $2v$
- Let the mass of car Y = m
    - Kinetic energy (KE) is equal to $E_k = \frac{1}{2}mv^2$
- KE of car X = $\frac{1}{2}2mv^2 = mv^2$
- KE of car Y = $\frac{1}{2}m2v^2 = 2mv^2$
    - Ratio $\frac{KE\ \ of\ \ car\ \ X}{KE\ \ of\ \ car\ \ Y} = \frac{mv^2}{2mv^2} = \frac{1}{2}$
- Therefore, car X has half the kinetic energy of car Y

Question 19 Medium [1 mark]

A trolley rolls along a horizontal surface and then travels up a slope before reaching a second horizontal surface. The slope is of length *L*. The trolley has mass *M*. The slope is at an angle *α* to the horizontal surface.

The second horizontal surface is at height *h* above the first surface.



Assume negligible frictional forces. The acceleration of free fall is known.

In order to determine the minimum initial velocity of the trolley for it to reach the top of the slope, which additional values are needed?

A *h* and *M*

B *M*, *L* and *h*

C *α*, *L* and *M*

D *h* only

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- At the bottom, the trolley needs to have a minimum amount of kinetic energy (KE) as a result of its initial velocity
- As the trolleys moves up the slope, it gains gravitational potential energy (GPE)
- From the conservation of energy:
    - KE at bottom = GPE at top + Work done against friction
- But since the question tells us that frictional forces are negligible, the work done against resistive forces is zero
- So, KE at bottom = GPE at top

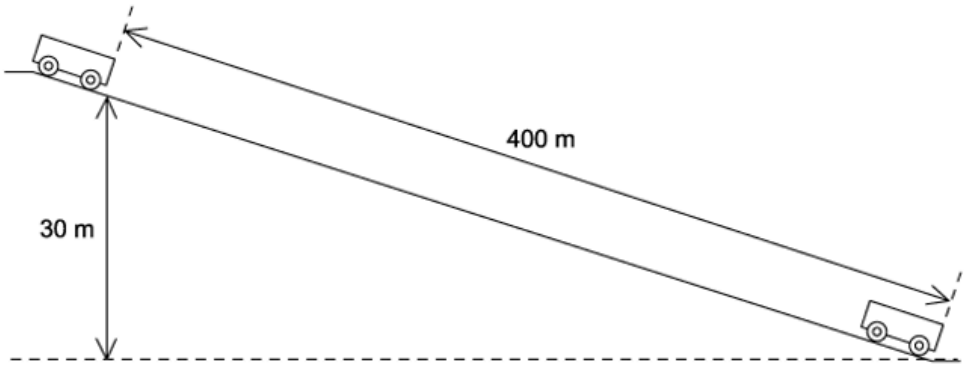
- $\frac{1}{2}mv^2 = mg\Delta h$

- The mass *m* can be eliminated on both sides leaving us with
  - $V^2 = 2g\Delta h$
- Therefore, we can see from this equation, to obtain the speed *v* we only need the values of *g* and *h*
- Since *g* is already known, so we only need to know the value of *h*

Question 20 Medium [1 mark]

A truck of mass 500 kg moves from rest at the top of a section of track 400 m long and 30 m high, as shown.

The frictional force acting on the truck is 250 N throughout its journey



What is the final speed of the truck?

A 14 m s^{-1}

B 24 m s^{-1}

C 31 m s^{-1}

D 190 m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- From the conservation of energy:
  - GPE of truck at top = KE of truck at bottom + Work done against friction
- GPE of truck at top $\Delta E_p = mg\Delta h = 500 \times 9.81 \times 30 = 147\ 150$ J
- Work done against friction $W = Fs = 250 \times 400 = 100\ 000$ J
- KE of truck at bottom = GPE of truck at top – Work done against friction
  - $\frac{1}{2}mv^2 = 147\ 150 - 100\ 000 = 47150$
  - $v = \sqrt{\frac{2 \times 47\ 150}{500}} = 13.7$

- Therefore, the final speed is 14 m s^{-1}

Question 21 Medium [1 mark]

A steel beam with a rectangular cross-section has dimensions 0.40 m × 0.30 m × 8.0 m and a uniform density of 7.8 × 10^3 kg m^{-3}. The beam is raised from ground level to a height of 12 m during construction.

What is the change in gravitational potential energy of the beam?

A 220 kJ

B 440 kJ

C 880 kJ

D 1800 kJ

Mark Scheme & Explanation:

The correct answer is C

- The density of the beam is given as $\rho = 7.8 \times 10^3 \text{ kg m}^{-3}$
- The volume of the beam is

$$V = 0.40 \times 0.30 \times 8.0 = 0.96 \text{ m}^3$$

- Mass is calculated using

$$m = \rho \times V$$

- Therefore

$$m = 7.8 \times 10^3 \times 0.96 = 7488 \text{ kg}$$

- Change in gravitational potential energy is

$$\Delta E_p = mg\Delta h$$

- Where:

$$m = \rho V$$

$$\Delta E_p = \rho V g \Delta h$$

$$\Delta E_p = 7.8 \times 10^3 \, 0.40 \times 0.30 \times 8.0 \times 9.81 \times 12$$

$$\Delta E_p = 881 \text{ kJ}$$

A is incorrect as this underestimates the change in height.

B is incorrect as this underestimates the mass of the steel beam.

D is incorrect as this overestimates the change in height or uses incorrect density.

Question 22 Hard [1 mark]

A loaded aeroplane has a total mass of 1.2×10^5 kg while climbing after take-off. It climbs at an angle of 23° to the horizontal with a speed of 50 m s^{-1}.

What is the rate at which it is gaining potential energy at this time?

A 2.3×10^6 J s^{-1}

B 2.5×10^6 J s^{-1}

C 2.3×10^7 J s^{-1}

D 2.5×10^7 J s^{-1}

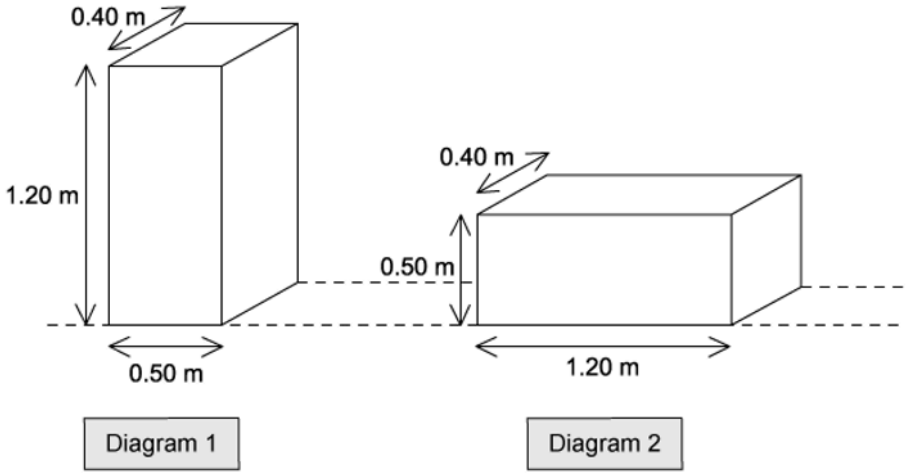
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Rate of gaining potential energy $= \frac{\Delta E_p}{\Delta t}$
- $\Delta E_p = mg\Delta h$
  - Where Δh is the vertical height
- So, to consider the gain in GPE, we need to consider the vertical component of speed
- The aeroplane climbs at a speed of 50 m s^{-1} at an angle of 23° to the horizontal
  - Vertical component of speed $= 50 \quad \sin \quad 23^\circ$
- Rate of gaining potential energy $= \frac{\Delta E_p}{\Delta t}$
- Rate of gaining potential energy $= \frac{mg\Delta h}{\Delta t} = mg\frac{\Delta h}{\Delta t}$
  - Where $\frac{\Delta h}{\Delta t}$ is the vertical speed at which the aeroplane climbs
- Rate of gaining potential energy $= 1.2 \times 10^5 \times 9.81 \times 50 \quad \sin \quad 23^\circ$
- Rate of gaining potential energy = 2.3×10^7 J s^{-1}

Question 23 Hard [1 mark]

A uniform solid cuboid of concrete of dimensions 0.50m × 1.20m × 0.40m and weight 4000 N rests on a flat surface with the 1.20 m edge vertical as shown in diagram 1.



What is the minimum energy required to roll the cuboid through 90° to the position shown in diagram 2 with the 0.50 m edge vertical?

A 200 J

B 400 J

C 1400 J

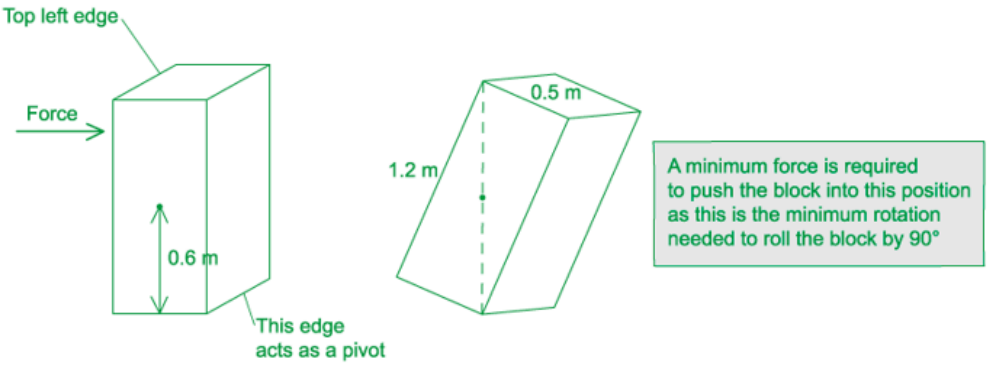
D 2600 J

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Since the solid cuboid is uniform, its centre of mass may be considered to be at the centre of the cuboid, that is, initially at a height of 0.6 m
  - Weight $= mg = 4000 \text{ N}$
- Therefore, before the force is applied, the GPE of the cuboid is equal to
  - $mgh = 4000 \times 0.6$
- As illustrated by the diagram, as the cuboid rolls about the bottom right edge, the centre of mass of the cuboid rises to a greater height, so, its potential energy increases and reaches a maximum and afterwards, the cuboid will fall under gravity

- So, the minimum energy is equal to the change in energy from these two positions
- Consider the position for maximum potential energy:
    - The centre of mass will be at the middle of the dotted line and this forms a right-angle triangle as shown in the diagram
- Using Pythagoras' theorem:
    - Vertical dotted line $= \sqrt{0.5^2 + 1.2^2} = 1.3 \quad \text{m}$
- Height of centre of mass in new position $= \frac{1.3}{2} = 0.65 \quad \text{m}$
- GPE of cuboid in new position :
    - $mgh = 4000 \times 0.65$
- Minimum energy required to roll cuboid = Rise in potential energy
- Minimum energy $= 4000 \times 0.65 - 4000 \times 0.6 = 200 \quad \text{J}$



Question 24 Hard [1 mark]

A steel sphere is dropped vertically onto a horizontal metal plate. The sphere hits the plate with a speed *u*, leaves it at a speed *v*, and rebounds vertically to half of its original height.

Which expression gives the value of $\frac{v}{u}$?

A $\frac{1}{2^2}$

B $\frac{1}{2}$

C $\frac{1}{\sqrt{2}}$

D $1 - \frac{1}{\sqrt{2}}$

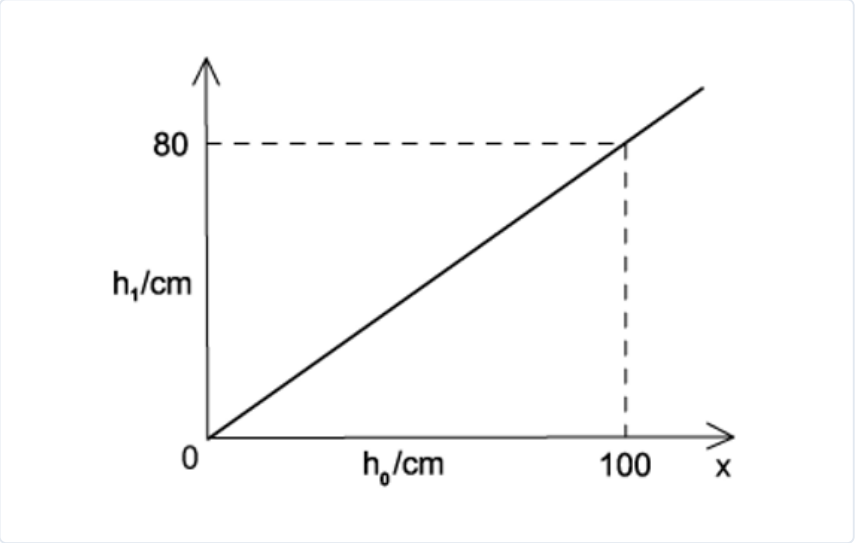
Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Let the height from which the sphere is dropped $= \Delta h$
- As the sphere falls, its gravitational potential energy gets converted into kinetic energy.
- Its speed when it hits the plate is *u*, so:
  - $mg\Delta h = \frac{1}{2}mu^2$
  - Speed $u = \sqrt{2g\Delta h}$
- As the sphere rebounds and move vertically up, its kinetic energy gets converted into gravitational potential energy
- The height is now half the height where it has dropped initially, so:
  - $\frac{1}{2}mv^2 = mg\frac{\Delta h}{2}$
  - Speed $v = \sqrt{g\Delta h}$
- So, the ratio of $\frac{v}{u}$ is:
  - Ratio $= \frac{v}{u} = \frac{\sqrt{g\Delta h}}{\sqrt{2g\Delta h}}$
  - Ratio $= \frac{v}{u} = \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{g\Delta h}}{\sqrt{g\Delta h}}$
  - Ratio $= \frac{v}{u} = \frac{1}{\sqrt{2}}$

Question 25 Hard [1 mark]

A ball released from a height h_0 above a horizontal surface rebounds to a height h_1 after one bounce. The graph that relates h_0 to h_1 is shown below.



If the ball (of mass m) was dropped from an initial height h and made three bounces, the kinetic energy of the ball immediately after the third impact with the surface was

A $0.8^3 mgh$

B $0.8^2 mgh$

C $1 - 3 \times 0.2 mgh$

D $1 - 0.8^3 mgh$

Mark Scheme & Explanation:

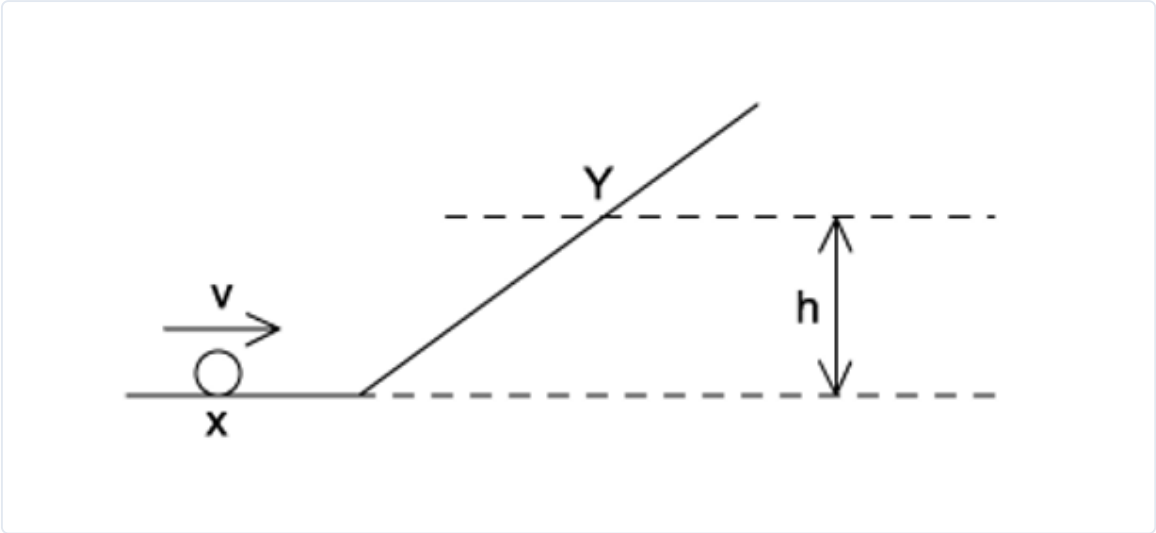
*The correct answer is **A** because:*

- Initially, the potential energy of ball $= mgh$
  - h_0 is the height before release and h_1 is the height after one bounce
- Since the graph is a straight line, the ratio of h_1 / h_0 (gradient) is constant
  - Gradient $= \frac{\Delta h_1}{\Delta h_0} = \frac{80 - 0}{100 - 0} = 0.8$
- So, after one bounce the potential energy becomes 0.8 of the initial potential energy

- The kinetic energy of the ball immediately after an impact with the surface is the same as the potential energy at maximum height as all kinetic energy is converted into potential energy
    - After 1st bounce, kinetic energy $= 0.8 \quad mgh$
    - After 2nd bounce, kinetic energy $= 0.8 \times 0.8mgh = 0.8^2 mgh$
    - After 3rd bounce, kinetic energy $= 0.8^3 mgh$

Question 26 Hard [1 mark]

An object of mass m passes a point X with a velocity v and slides up a frictionless incline to stop at point Y, which is at a height h above X.



A second object of mass $\frac{m}{2}$ passes X with a velocity of $\frac{v}{2}$.

What height will the second object reach?

A $\frac{h}{4}$

B $\frac{h}{2}$

C $\frac{h}{\sqrt{2}}$

D $h\sqrt{2}$

Mark Scheme & Explanation:

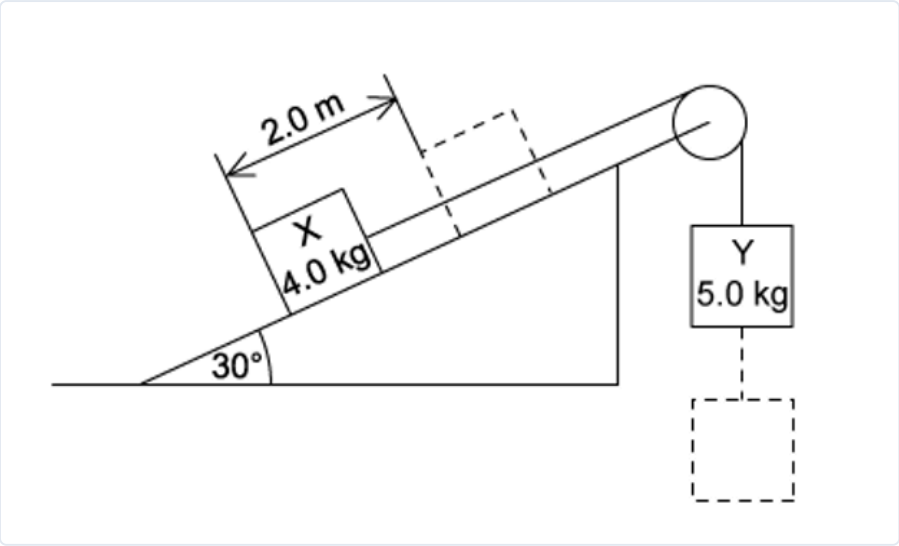
*The correct answer is **A** because:*

- For the object of mass *m* and velocity *v*:
  - At point X, Kinetic energy = $\frac{1}{2}mv^2$
- From conservation of energy :
- Potential energy at point Y = Kinetic energy at point X
  - $mgh = \frac{1}{2}mv^2$
  - Height, $h = \frac{v^2}{2g}$

- For the object of mass $\frac{m}{2}$ and velocity $\frac{v}{2}$:
    - At point X, Kinetic energy $= \frac{1}{2}\frac{m}{2}\frac{v}{2}^2 = \frac{mv^2}{16}$
- Potential energy at point Y = Kinetic energy at point X
- Let the height at point Y = x
    - $\frac{m}{2} \quad gx = \frac{mv^2}{16}$
    - Height, $x = \frac{v^2}{8g} = \frac{1}{4}\frac{v^2}{2g} = \frac{h}{4}$

Question 27 Hard [1 mark]

The diagram shows two bodies X and Y connected by a light cord passing over a light, free-running pulley. X starts from rest and moves on a smooth plane inclined at 30° to the horizontal.



What will be the total kinetic energy of the system when X has travelled 2.0 m along the plane?

A 20 J

B 59 J

C 64 J

D 132 J

Mark Scheme & Explanation:

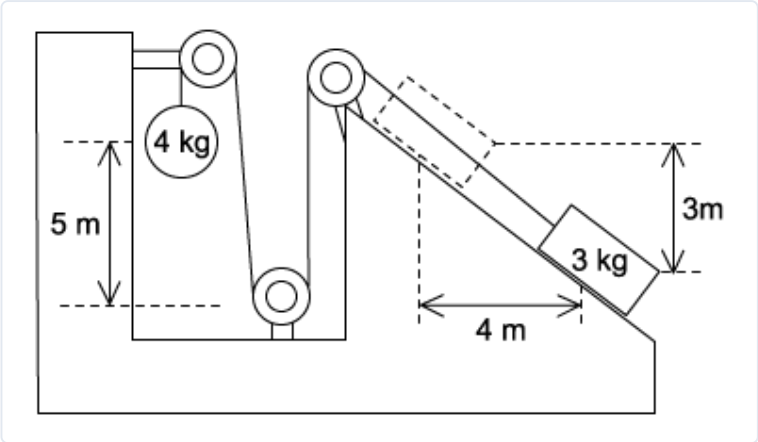
*The correct answer is **B** because:*

- The acceleration due to gravity of 9.8 m s^{-2} acts downwards on Y since it is hanging freely
  - Weight of Y = $mg = 5.0 \times 9.8 = 49$ N
- As X is attached to Y, it will move up the slope as Y falls
- So, the acceleration in the whole system is therefore **not** 9.8 m s^{-2}
- We need to find the resultant acceleration which acts on the total mass in the system. The component of the weight of X will act on the system:
  - $mg \sin 30 = 4.0 \times 9.8 \times \sin 30 = 19.6$ N

- The weight of Y contributes a force up the slope while the component of weight of X contributes a force down the slope
- Resultant force in system $= ma = 49 - 19.6 = 29.4 \quad \text{N}$
  - $4.0 + 5.0 \quad a = 29.4 \quad \text{N}$
  - Acceleration, $a = \frac{29.4}{9} = 3.27 \quad \text{m} \quad \text{s}^{-2}$
- This is the resultant acceleration on each X and Y.
- Initially both X and Y are at rest. Consider the equation for uniformly accelerated motion:
  - $v^2 = u^2 + 2as$ where u = 0, v = ?, s = 2.0 m and a = 3.27 m s^{-2}
  - $v^2 = 0 + 23.272$
  - $v = \sqrt{2 \times 3.27 \times 2} = 3.62 \quad \text{m} \quad \text{s}^{-1}$
- Kinetic energy of X $= \frac{1}{2} 4.03.62^2$
- Kinetic energy of Y $= \frac{1}{2} 5.03.62^2$
- Total kinetic energy of system $= \frac{1}{2} 4.03.62^2 + \frac{1}{2} 5.03.62^2 = 58.96 \quad \text{J}$
  - Therefore, the total kinetic energy is 59 J

Question 28 Hard [1 mark]

A 4 kg ball and a 3 kg block of wood were placed in an arrangement as shown below. When the ball was released, it fell through a distance of 5 m, pulling the block of wood up the ramp.



If the pulleys were well oiled and the surface of the ramp exerts a frictional force of 2 N, what is the final speed of the sphere after falling through the 5 m?

A 5.0 m s^{-1}

B 5.3 m s^{-1}

C 6.9 m s^{-1}

D 10.0 m s^{-1}

Mark Scheme & Explanation:

*The correct answer is **B** because:*

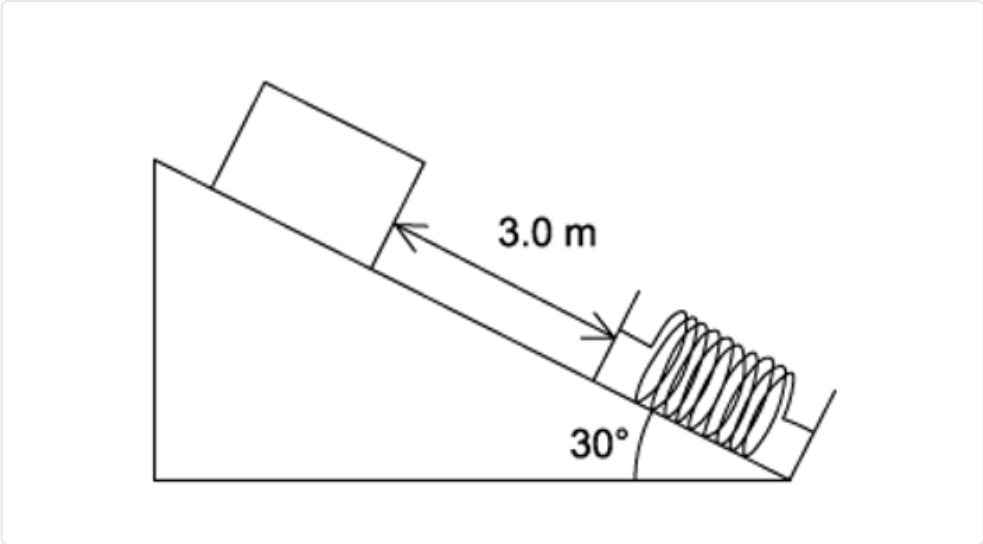
- From the conservation of energy:
  - Change in GPE = Change in KE + Work done against friction
- In equation form we can write this such that:
  - $\Delta E_p = \Delta E_k + Fs$
- The change in GPE comes from the loss in GPE from the 4 kg mass and the gain in GPE from the 3 kg mass:
  - $\Delta E_p = mg\Delta h = 4 \times 9.8 \times 5 - 3 \times 9.8 \times 3 = 107.8$ J
- The change in KE comes from the gain in GPE from the 4 kg mass and the 3 kg mass, the masses are connected so both travel with speed *v*:

- $\Delta E_k = \frac{1}{2}mv^2 = \frac{1}{2}3v^2 + \frac{1}{2}4v^2 = \frac{7v^2}{2}$

- The work done against friction is given by $Fs = 2 \times 5 = 10 \quad \text{J}$
  - $107.8 = \frac{7v^2}{2} + 10$
  - $v^2 = \frac{2}{7}107.8 - 10 = 27.9$
  - $v = \sqrt{27.9} = 5.29 \quad \text{m} \quad \text{s}^{-1}$
- Therefore, the final speed of the mass is 5.3 m s^{-1}

Question 29 Hard [1 mark]

A block of mass 12 kg is released from rest and it slides down a 30° frictionless incline plane. It is stopped by a spring with a spring constant *k* as shown below.



The initial distance between the point of release and the uncompressed spring is 3.0 m. After the block comes to rest momentarily, the spring is compressed by 1.3 m.

What is the value of *k*?

A 200 N m^{-1}

B 250 N m^{-1}

C 300 N m^{-1}

D 350 N m^{-1}

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- From the conservation of energy:
  - Change in GPE = Change in elastic potential energy + Work done against friction
- We are told the slope is frictionless, so we can disregard the work done against friction as it is negligible
- Change in GPE as the block slides down the slope and compresses the spring is
  - $\Delta E_p = mg\Delta h = 129.83 + 1.3 \quad \sin \quad 30 = 252.84 \quad \text{J}$

- The loss in GPE is equal to the elastic energy stored by the spring when it is compressed, so:
  - $\Delta E_p = \frac{1}{2}kx^2 = \frac{1}{2}k1.3^2$
- Where *k* is the spring constant and *x* is the compression of the spring
  - $252.84 = \frac{1}{2}k1.3^2$
  - $k = \frac{2\times 252.84}{1.3^2} = 299.2 \quad \text{N} \quad \text{m}^{-1}$
- Therefore, the spring constant is 300 N m^{-1}

Question 30 Hard [1 mark]

A falling mass pulls a trolley from rest on a horizontal table. Given that:

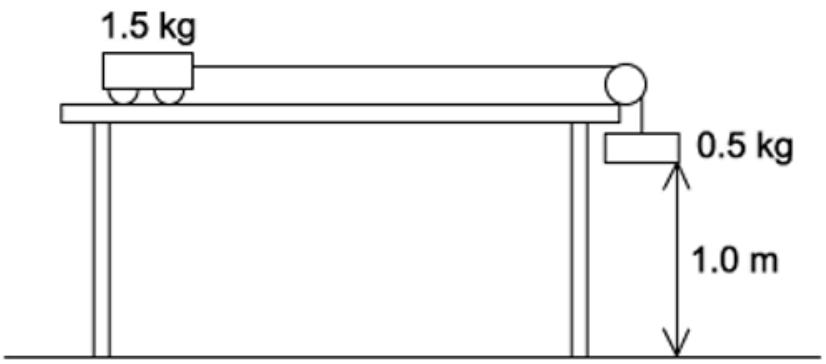
The mass of the trolley = 1.5 kg

The mass of the falling mass = 0.5 kg

The distance the mass falls through = 1.0 m

Frictional force from the table = 0.8 N

The distance the trolley moves = 1.2 m



What is the maximum kinetic energy of the trolley?

A 3.0 J

B 4.9 J

C 15 J

D 20 J

Mark Scheme & Explanation:

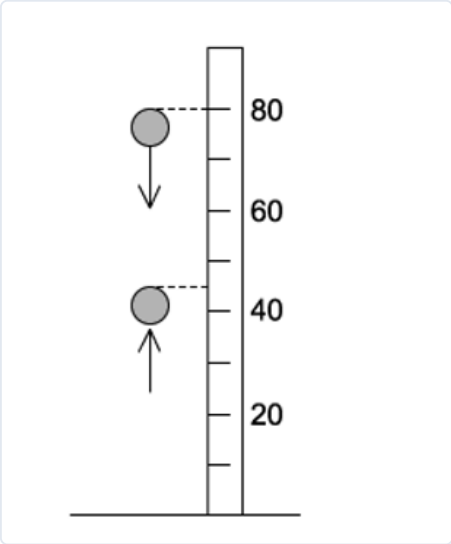
*The correct answer is **A** because:*

- From the conservation of energy:
  - Loss in PE of the falling mass = Gain in KE of the whole system + Work done against friction
- Loss in PE is equal to $= m_m g \Delta h = 0.5 \times 9.8 \times 1 = 4.9$ J

- Total gain in KE is equal to $= \frac{1}{2}m_t + m_m v^2 = \frac{1}{2}1.5 + 0.5v^2 = v^2$
    - Where m_t and m_m are the masses of the trolley and falling mass respectively
- Work done against friction = $Fs = 0.8 \times 1.2 = 0.96 \quad \text{J}$
- The final speed of the trolley and mass is:
    - $4.9 = v^2 + 0.96$
    - $v = \sqrt{4.9 - 0.96} = 1.98 \quad \text{m} \quad \text{s}^{-1}$
- So, the maximum kinetic energy of the trolley is equal to
    - $\frac{1}{2}m_t v^2 = \frac{1}{2}1.51.98^2 = 2.955 \quad \text{J}$
- Therefore, the maximum kinetic energy of the trolley is 3.0 J

Question 31 Hard [1 mark]

A solid rubber ball has a diameter of 8.0 cm. It is released from rest with the top of the ball 80 cm above a horizontal surface. It falls vertically and then bounces back up so that the maximum height reached by the top of the ball is 45 cm, as shown.



If the kinetic energy of the ball is 0.75 J just before it strikes the surface, what is its kinetic energy just after it leaves the surface?

A 0.36 J

B 0.39 J

C 0.40 J

D 0.42 J

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Let's first consider the bottom of the ball at 72 cm at the start and 37 cm at the end
- From the conservation of energy, as the ball falls, all the potential energy (PE) is being converted to kinetic energy
- Just before it strikes the surface, the kinetic energy is 0.75 J as all PE has been converted to KE
- At this point, the bottom of the ball has zero PE and a KE of 0.75 J
    - This value of KE is the value at the bottom of the ball, so, throughout our calculation we need to consider the bottom of the ball instead of the top

- PE of bottom of the ball at the point of release = KE of bottom of ball just before striking the surface
  - $\Delta E_p = mg\Delta h = \Delta E_k$
  - $m \times 9.81 \times 0.72 = 0.75$
  - Mass of the ball, $m = \frac{0.75}{9.81 \times 0.72} = 0.1062 \quad \text{kg}$
- Now, for the rebound, all of the kinetic energy just after the ball leaves the surface is converted to potential energy at the maximum height reached
- The maximum of height of the bottom of the ball after rebound is 45 – 8 = 37 cm
- KE just after leaving the surface = PE of bottom of ball at a height of 37 cm
  - $mg\Delta h = \Delta E_k$
  - $E_k = 0.1062 \times 9.81 \times 0.37 = 0.3855 \quad \text{J}$
- Therefore, the kinetic energy just after the ball leaves the surface is 0.39 J
  - Note that 0.75 J is the KE of the ball just before hitting the surface while 0.39 J is the KE of the ball just after leaving the surface. This indicates that some energy (0.75 – 0.39) J has been lost